

FPGA100T / FULL-SYSTEM DEVELOPMENT

CORE +116 FPGA-LINKED SIGNALS +1 ANALOG CONTACT / DRAFT

Goal: power the FPGA, program it through the shared cable, and boot from flash.

HOW THE CORE WORKS



J1, J2, J3, all test points, reset button and LED removed.

Arrows above explain functions; electrical connections are on the linked pages.

WHAT IS VERIFIED

116 FPGA-linked signals +1 analog-reserved contact. See current native netlist and DRC for actual connectivity.

This does not prove power quality, routability, timing closure or hardware operation.

SCHEMATIC PAGES

02	FPGA banks 14 and 15
03	FPGA banks 16 and 34
04	FPGA bank 35
05	FPGA configuration and analog
06	FPGA power and ground
07	Cable input capacitors / 12 V draft
08	Core supply / about 1.0 V
09	Auxiliary supply / 1.8 V
10	Bank16 link supply / 2.5 V
11	ASIC bank supply / 1.5 V
12	Core and BRAM capacitors
13	Auxiliary and ADC capacitors
14	Configuration / link bank capacitors
15	1.5 V bank capacitors
16	Flash memory
17	JTAG and fixed mode straps
18	Reference clock
19	Electrical—rule power markers
20	J4 / shared power and JTAG connector

116 FPGA + 1 ANALOG SIGNAL

116 FPGA-linked signals +1 analog contact on sheet21; J6.58-60 spare.

XEM8310 receiver; cable data protocol remains open.

ENGINEERING DRAFT – NOT FOR FABRICATION

Sheet: /
File: FPGA100T_33x36_Routing.kicad_sch

Title: FPGA100T core / power, boot, clock and JTAG

Size: A3 Date: 2026-09-26

Rev: AC_IN ANALOG CORRECT

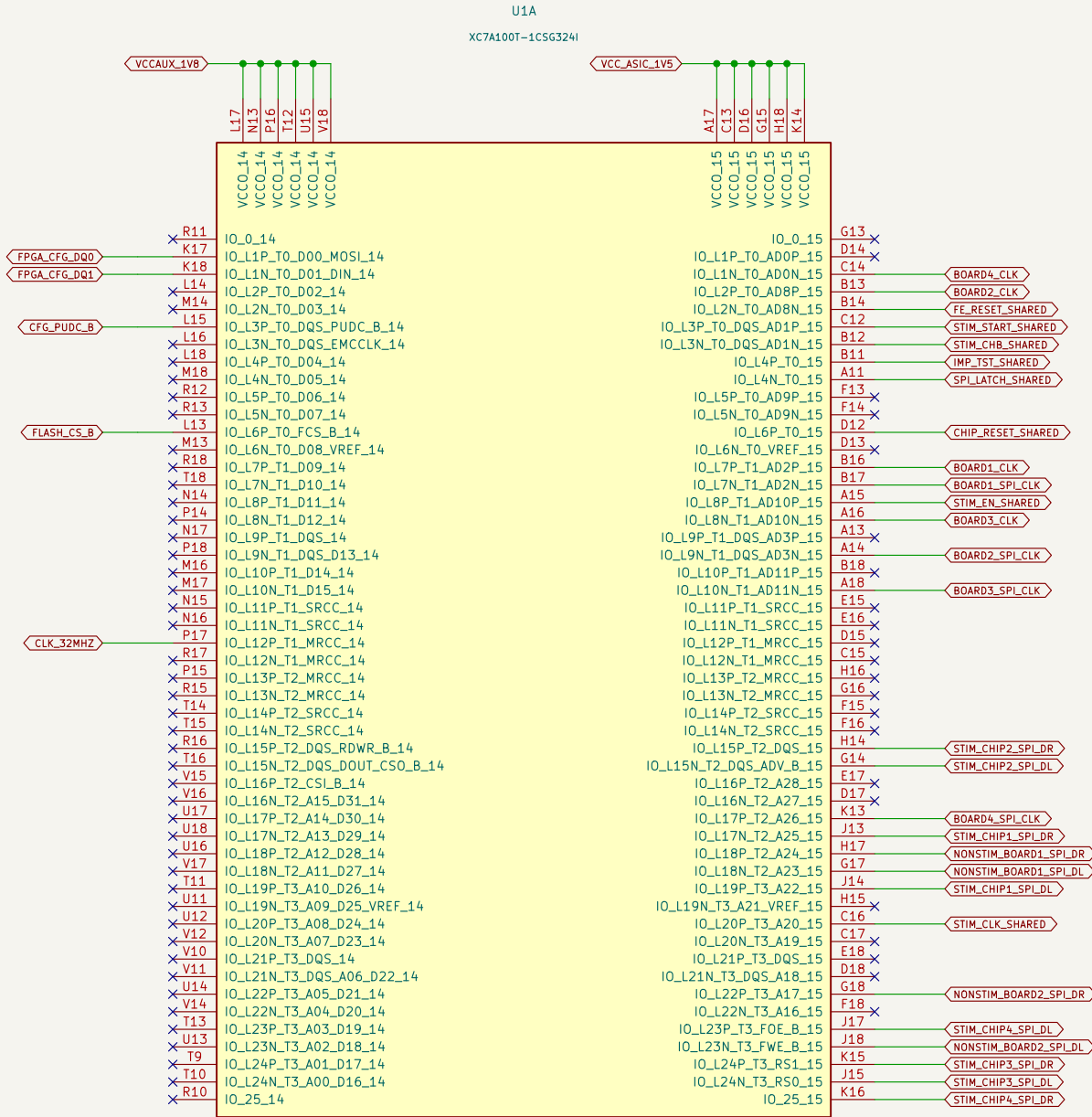
SIZE: A3	
KiCad E.D.A. 10.0.6	

Id: 1/21

U1 / XC7A100T-CSG324 / I/O banks 14 and 15

116 FPGA-linked signals +1 analog contact. Bank15/34/35 digital limits and clock/timing remain under review.
AC_IN is analog0–1.5 V: J5.46 only, FPGA A13 NC. IMP_TST: assumed0/1.5V digital test control; polarity and inactive state still pending.

A13 intentionally NC. AC_IN is analog0–1.5 V and is not an FPGA GPIO.



ENGINEERING DRAFT – NOT FOR FABRICATION

Howard Wang / XC7A100T CSG324

Sheet: /U1 banks 14 and 15/

File: fpga_unit_01.kicad_sch

Title: U1 / XC7A100T-CSG324 / I/O banks 14 and 15

Size: A3 Date: 2026-09-26

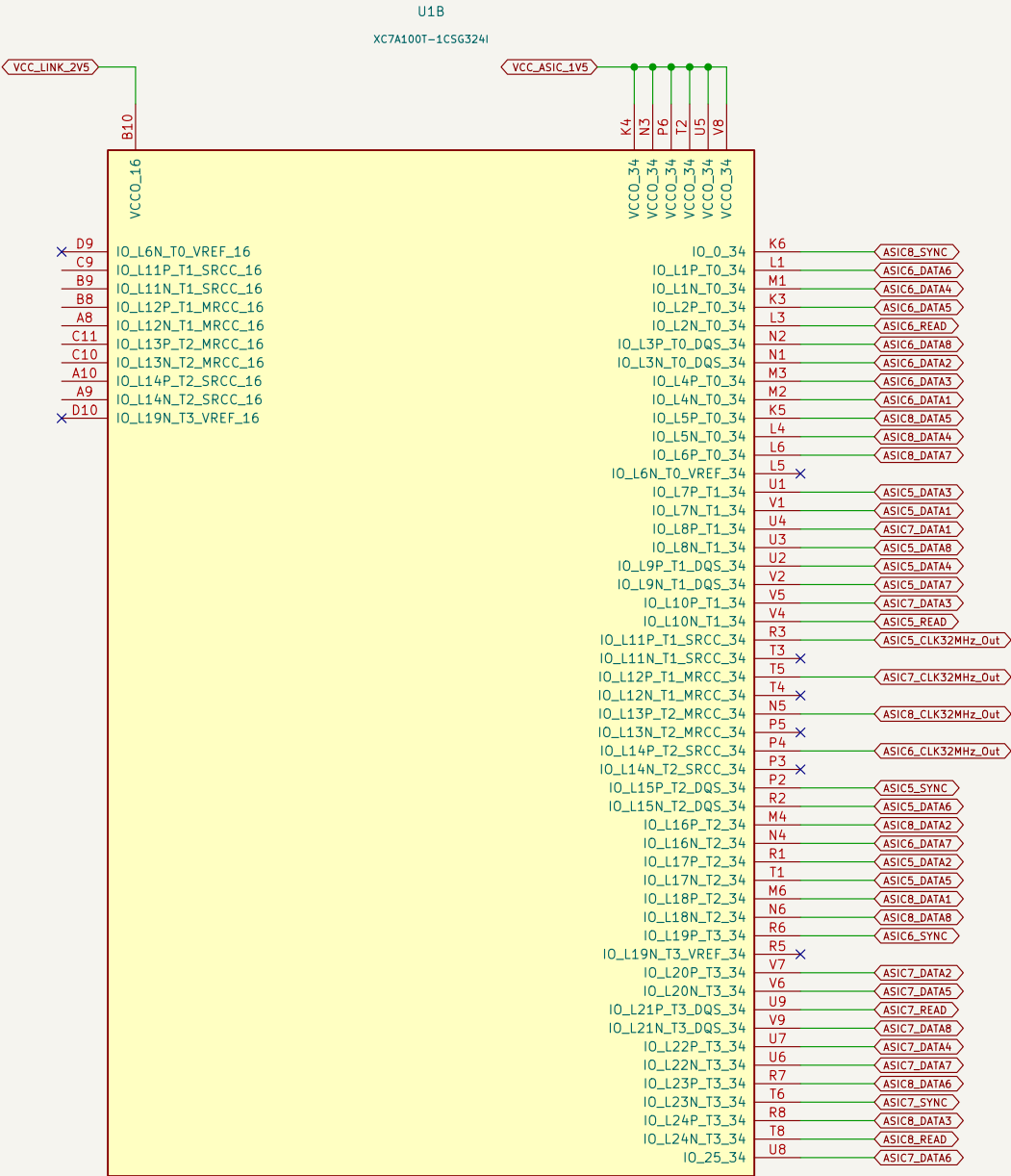
KiCad E.D.A. 10.0.6

Rev: AC_IN ANALOG CORRECTI

Id: 2/21

U1 / XC7A100T-CSG324 / I/O banks 16 and 34

116 FPGA-linked signals +1 analog contact. Bank15/34/35 digital limits and clock/timing remain under review.
AC_IN is analog0-1.5 V: J5.46 only, FPGA A13 NC. IMP_TST: assumed0/1.5V digital test control; polarity and inactive state still pending.



ENGINEERING DRAFT – NOT FOR FABRICATION

Howard Wang / XC7A100T CSG324

Sheet: /U1 banks 16 and 34/

File: fpga_unit_02.kicad_sch

Title: U1 / XC7A100T-CSG324 / I/O banks 16 and 34

Size: A3

Date: 2026-09-26

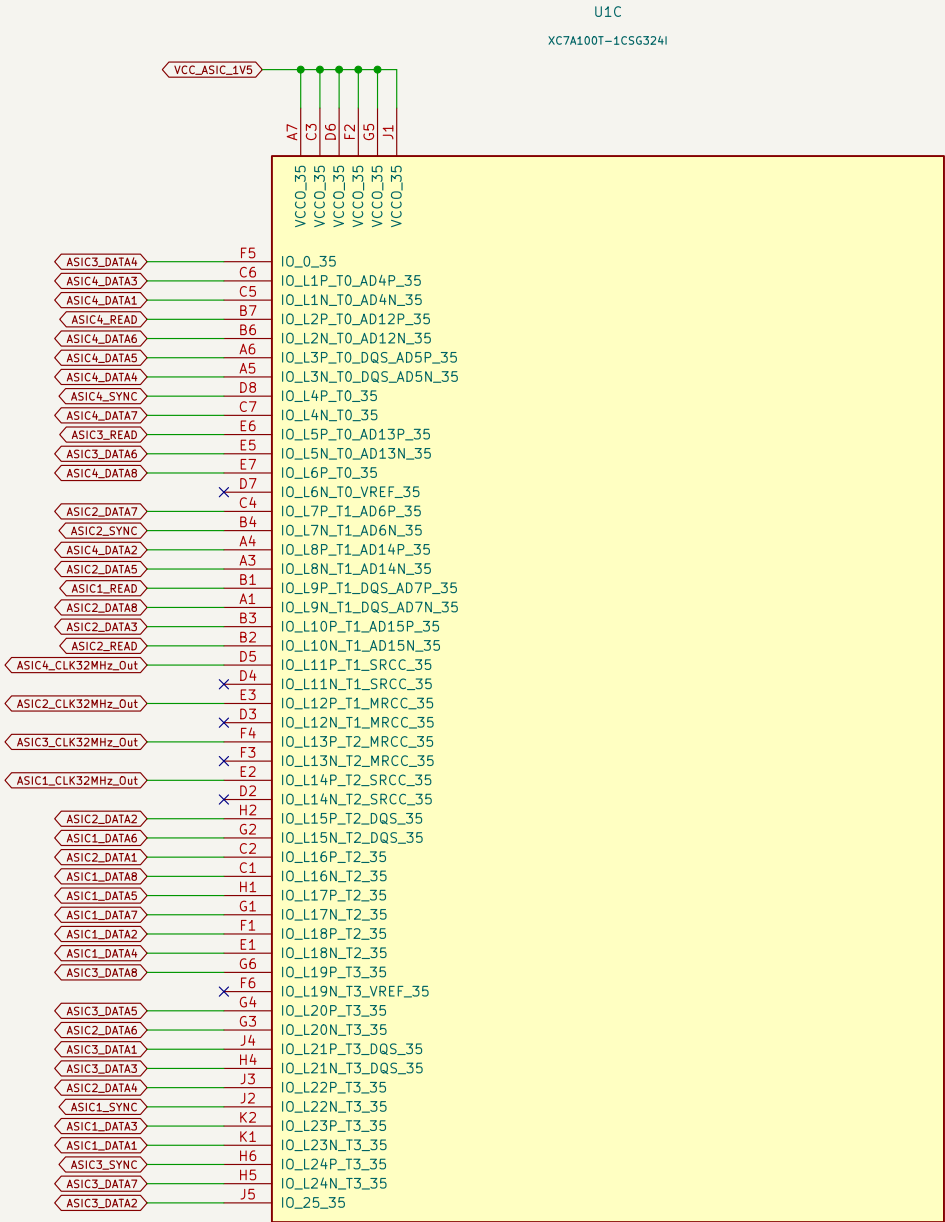
Rev: AC_IN ANALOG CORRECTI

KiCad E.D.A. 10.0.6

Id: 3/21

U1 / XC7A100T-CSG324 / I/O bank 35

116 FPGA-linked signals +1 analog contact. Bank15/34/35 digital limits and clock/timing remain under review.
AC_IN is analog0-1.5 V: J5.46 only, FPGA A13 NC. IMP_TST: assumed0/1.5V digital test control; polarity and inactive state still pending.



ENGINEERING DRAFT – NOT FOR FABRICATION
Howard Wang / XC7A100T CSG324
Sheet: /U1 bank 35/
File: fpga_unit_03.kicad_sch

Title: U1 / XC7A100T-CSG324 / I/O bank 35

Size: A3
KiCad E.D.A. 10.0.6

Date: 2026-09-26

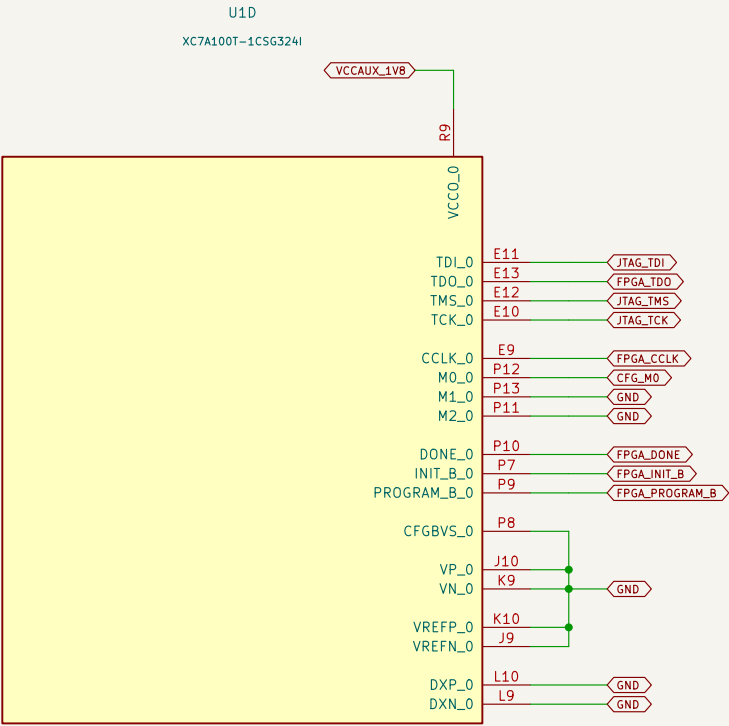
Rev: AC_IN ANALOG

Id: 4/21

CORRECTION

U1 / XC7A100T–CSG324 / Configuration, JTAG and analog support

15 x 15 mm CSG324. The117 allocated ASIC contacts comprise116 FPGA–linked signals and1 analog contact. Unassigned FPGA pins are deliberately open; interface details remain under review.

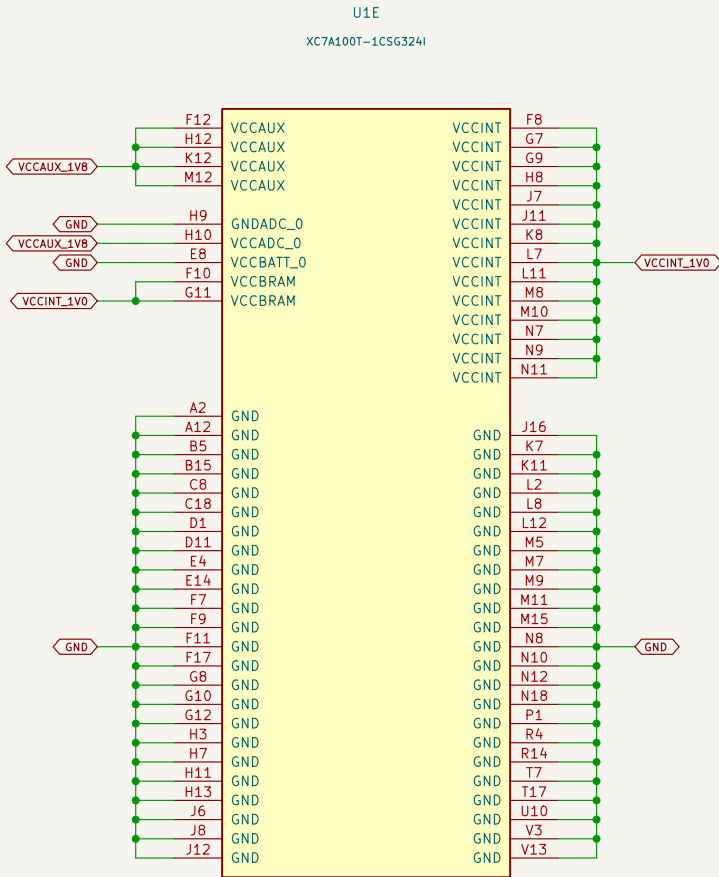


Required–ground correction: unused DXP/DXN grounded per AMD UG475 Table1–12.

ENGINEERING DRAFT – NOT FOR FABRICATION		
Howard Wang / XC7A100T CSG324		
Sheet: /U1 configuration, JTAG and analog/ File: fpga_unit_04.kicad_sch		
Title: U1 / XC7A100T–CSG324 / Configuration, JTAG and...		
Size: A3	Date: 2026–09–26	Rev: FIXED SPI M1 TO GND
KiCad E.D.A. 10.0.6		Id: 5/21

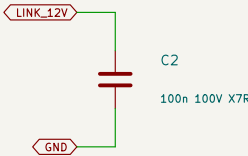
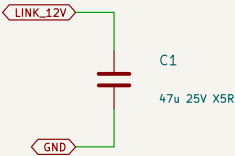
U1 / XC7A100T–CSG324 / Core, auxiliary power and ground

15 x 15 mm CSG324. The117 allocated ASIC contacts comprise116 FPGA–linked signals and1 analog contact. Unassigned FPGA pins are deliberately open; interface details remain under review.



Cable input capacitors

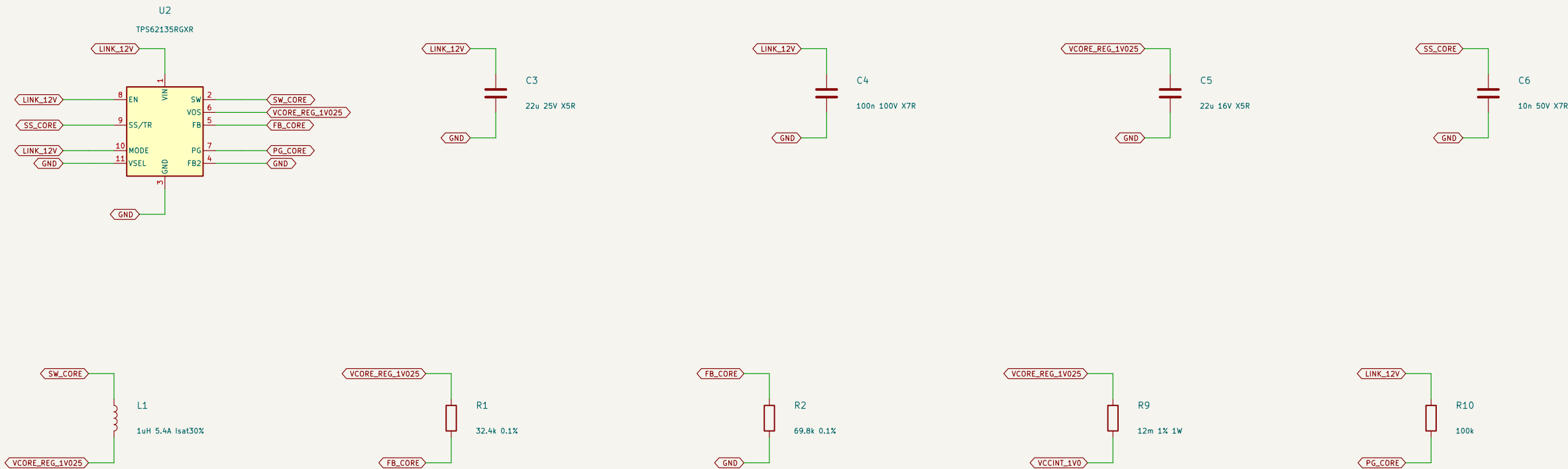
Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



ENGINEERING DRAFT – NOT FOR FABRICATION		
Howard Wang / XC7A100T CSG324		
Sheet: /Cable input capacitors/ File: core_01_01.kicad_sch		
Title: Cable input capacitors		
Size: A3	Date: 2026-09-26	Rev: DEV 0.4
KiCad E.D.A. 10.0.6	Id: 7/21	

Core regulator / 1.0 V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



Required-ground correction: FB2-grounded for single-divider/VSEL-low mode, TI TPS62135 RevB Fig5.

ENGINEERING DRAFT – NOT FOR FABRICATION

Howard Wang / XC7A100T CSG324

Sheet: /Core regulator / 1.0 V/
File: core_02_01.kicad_sch

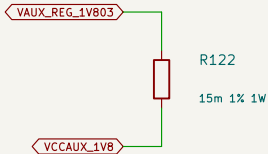
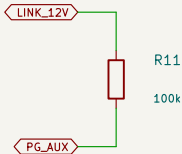
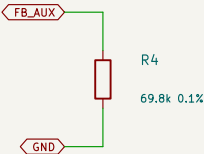
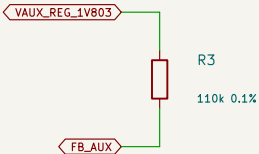
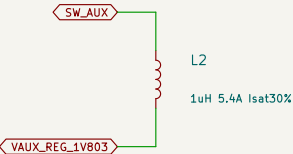
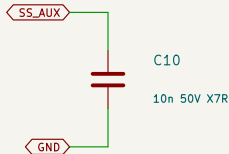
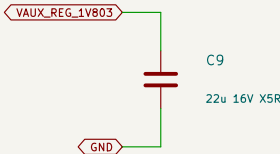
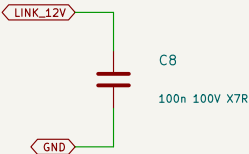
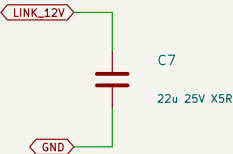
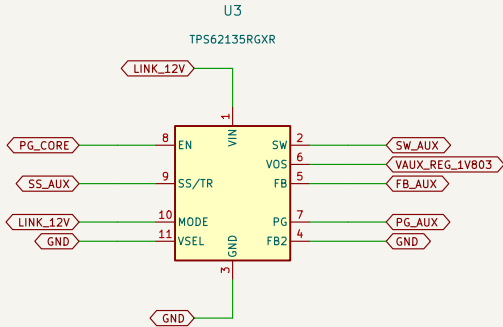
Title: Core regulator / 1.0 V

Size: A3
Date: 2026-09-26
KiCad E.D.A. 10.0.6

Rev: ASIC117 + REQUIRED GND
Id: 8/21

Auxiliary regulator / 1.8 V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.

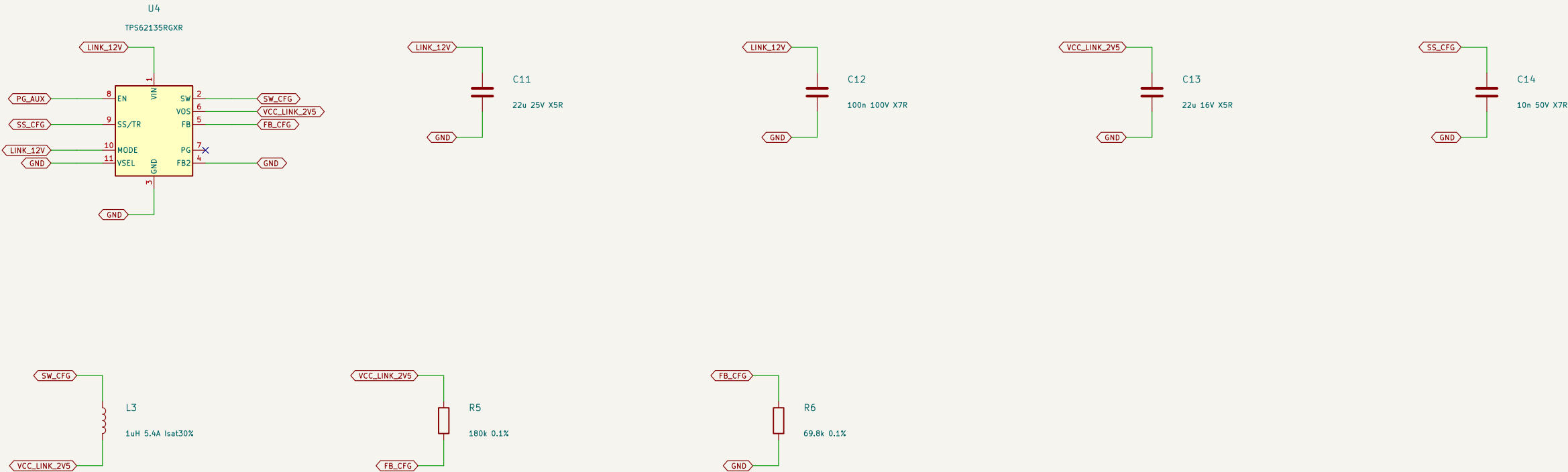


Required-ground correction: FB2 grounded for single-divider/VSEL-low mode, TI TPS62135 RevB Fig5.

ENGINEERING DRAFT – NOT FOR FABRICATION		
Howard Wang / XC7A100T CSG324		
Sheet: /Auxiliary regulator / 1.8 V/ File: core_03_01.kicad_sch		
Title: Auxiliary regulator / 1.8 V		
Size: A3	Date: 2026-09-26	Rev: ASIC117 + REQUIRED GND
KiCad E.D.A. 10.0.6		Id: 9/21

Bank16 link regulator / 2.5 V

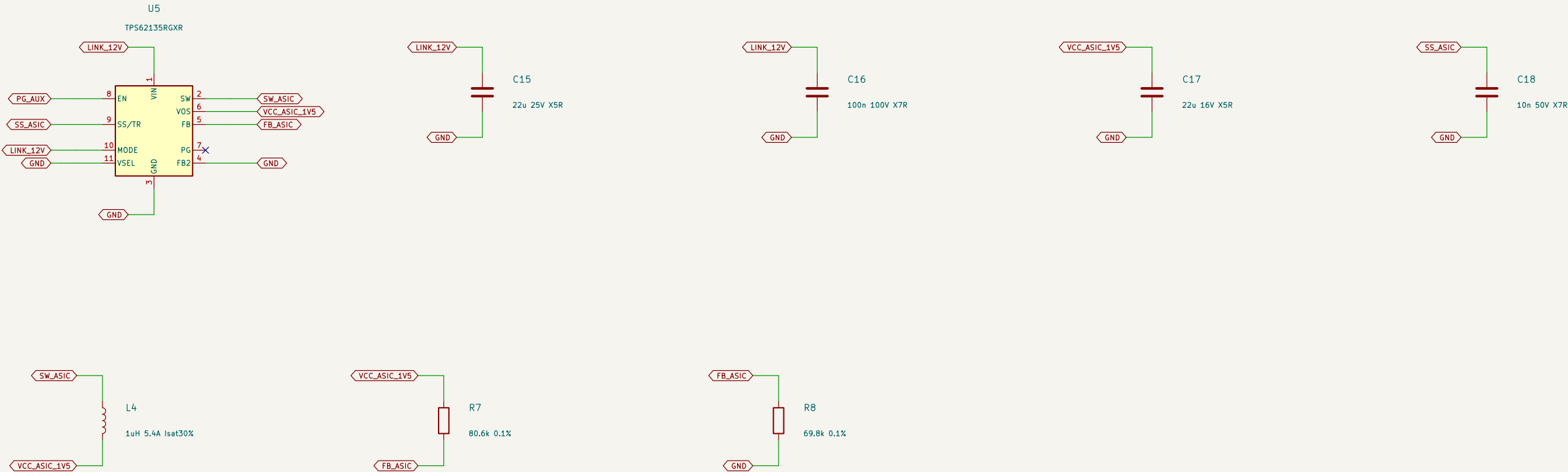
Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



Required-ground correction: FB2 grounded for single-divider/VSEL-low mode, TI TPS62135 RevB Fig5.

ASIC bank regulator / 1.5 V

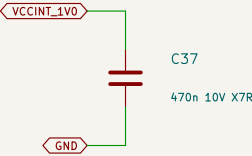
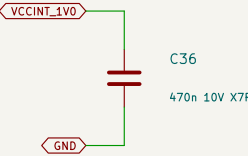
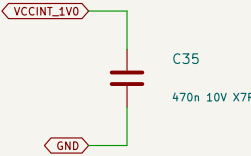
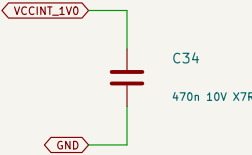
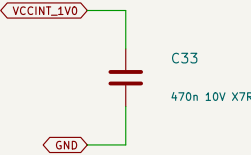
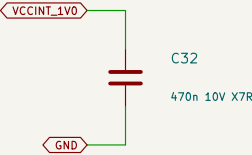
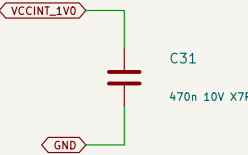
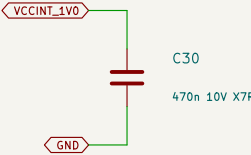
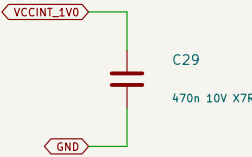
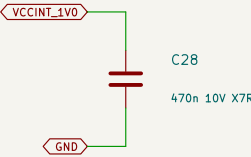
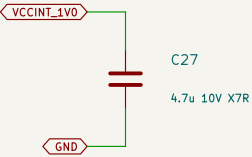
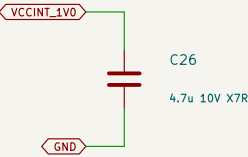
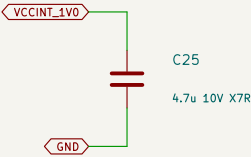
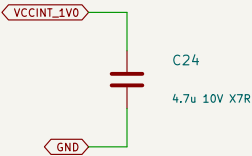
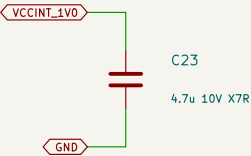
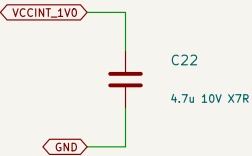
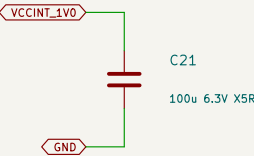
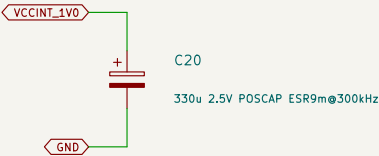
Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



Required-ground correction: FB2 grounded for single-divider/VSEL-low mode, TI TPS62135 RevB Fig5.

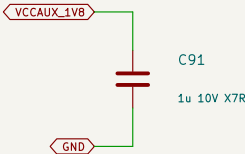
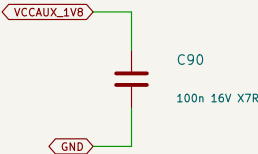
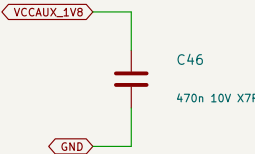
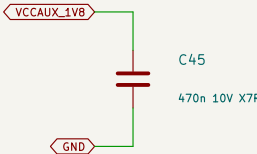
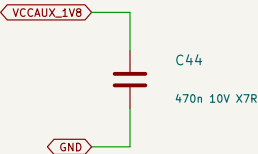
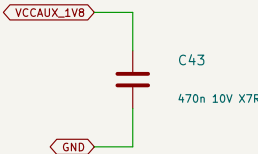
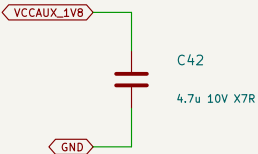
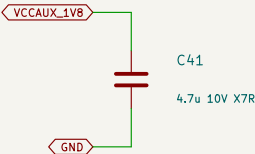
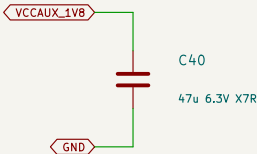
Decoupling / core and BRAM / 1.0 V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



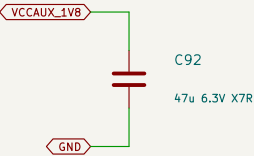
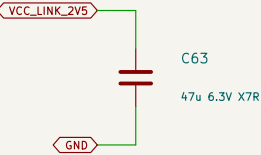
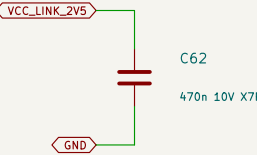
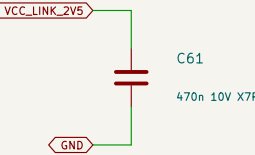
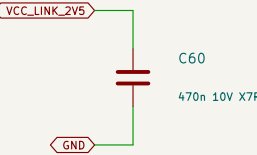
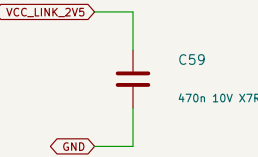
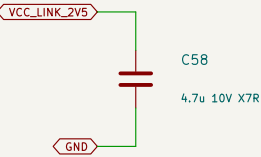
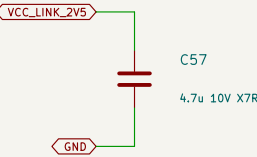
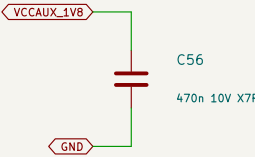
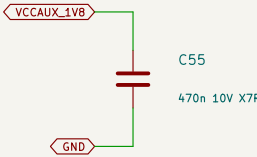
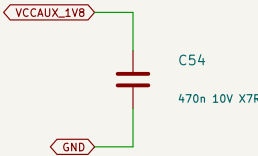
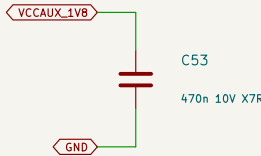
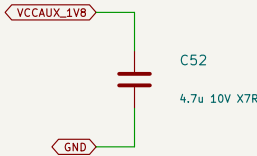
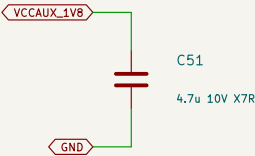
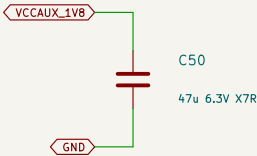
Decoupling / auxiliary and XADC / 1.8 V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



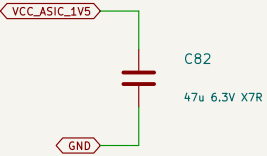
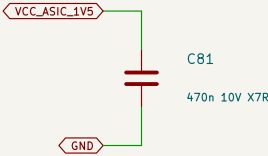
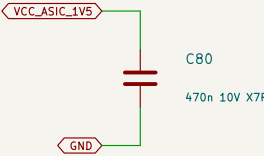
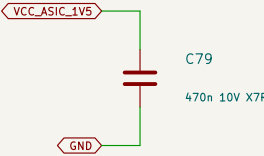
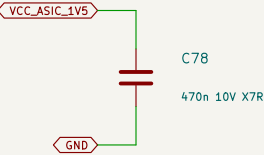
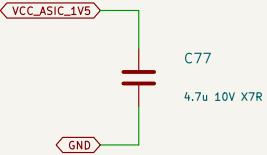
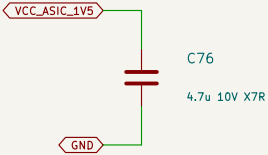
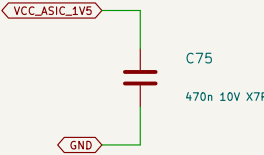
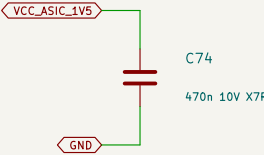
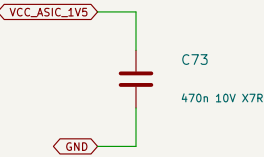
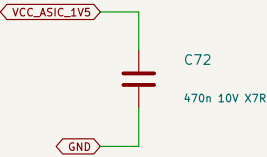
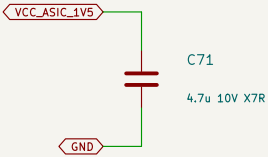
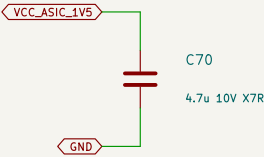
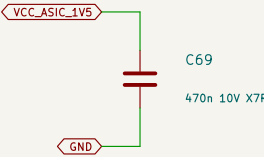
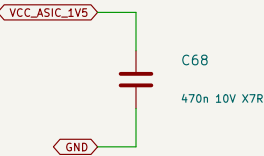
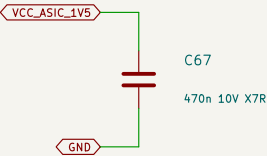
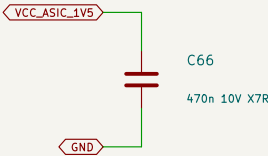
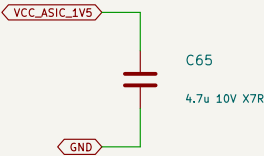
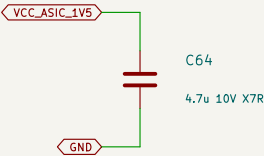
Decoupling / config1.8V and link2.5V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



Decoupling / ASIC-facing banks / 1.5 V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



ENGINEERING DRAFT – NOT FOR FABRICATION

Howard Wang / XC7A100T CSG324

Sheet: /Decoupling / ASIC-facing banks / 1.5 V/
File: core_09_01.kicad_sch

Title: Decoupling / ASIC-facing banks / 1.5 V

Size: A3 Date: 2026-09-26

KiCad E.D.A. 10.0.6

Rev: DEV 0.4

Id: 15/21

Boot flash

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.

A

B

C

D

E

F

A

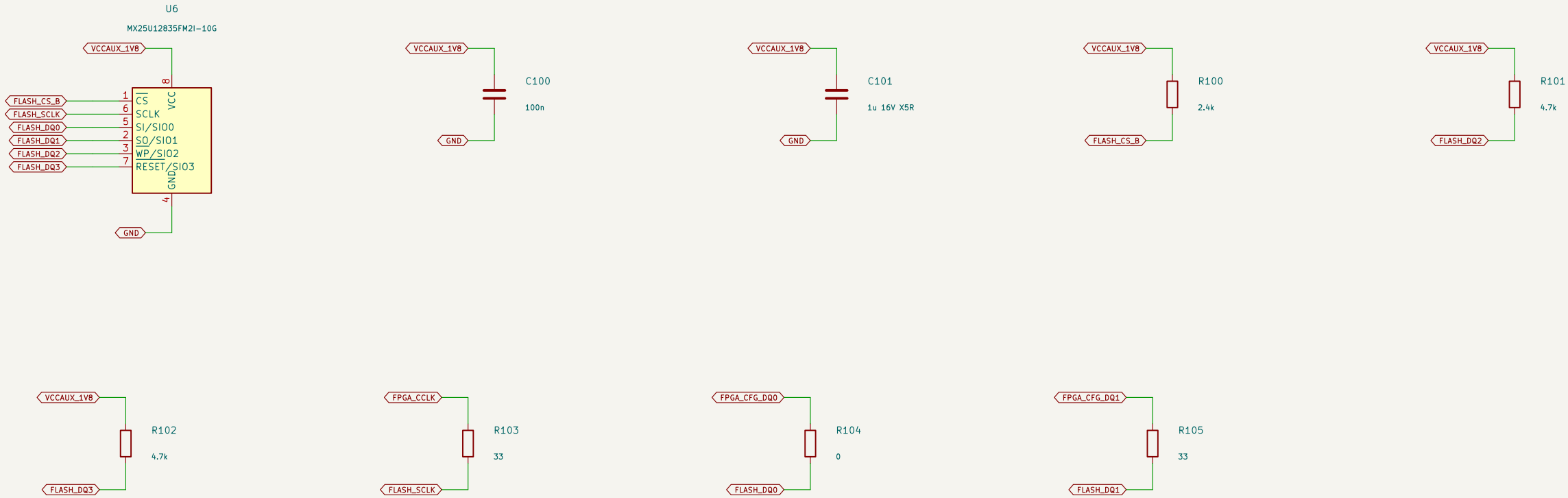
B

C

D

E

F



ENGINEERING DRAFT – NOT FOR FABRICATION

Howard Wang / XC7A100T CSG324

Sheet: /Boot flash/
File: core_10_01.kicad_sch

Title: Boot flash

Size: A3 Date: 2026-09-26

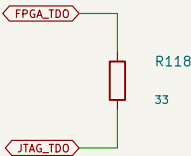
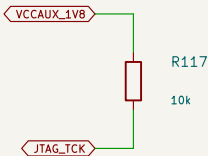
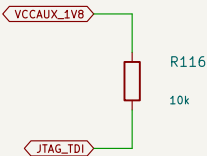
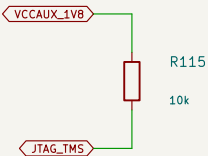
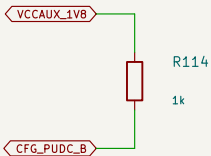
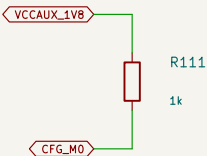
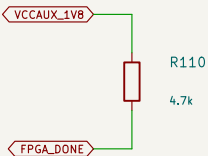
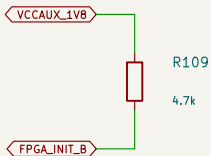
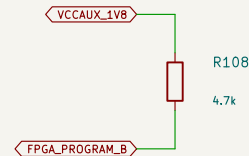
KiCad E.D.A. 10.0.6

Rev: DEV 0.4

Id: 16/21

Programming JTAG

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.

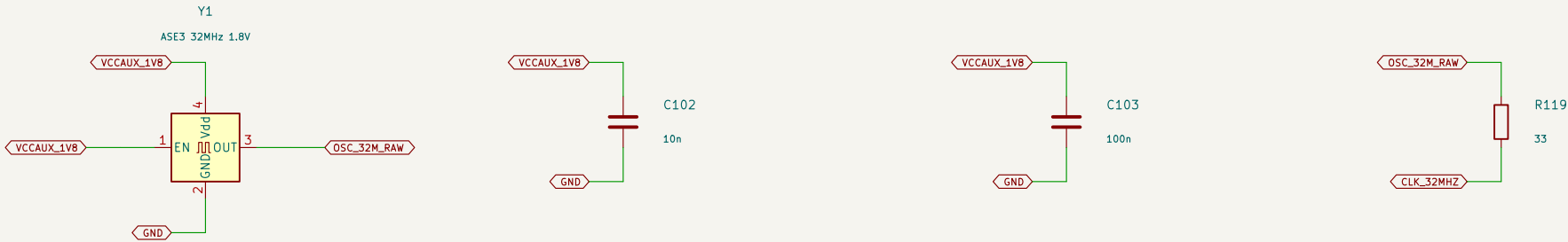


R112 removed: M1/P13 now directly grounded for fixed Master SPI001; JTAG remains available (AMD UG470 v1.17 pp7–8,17,20).

ENGINEERING DRAFT – NOT FOR FABRICATION		
Howard Wang / XC7A100T CSG324		
Sheet: /Programming JTAG/ File: core_11_01.kicad_sch		
Title: Programming JTAG		
Size: A3	Date: 2026–09–26	Rev: FIXED SPI M1 TO GND
KiCad E.D.A. 10.0.6		Id: 17/21

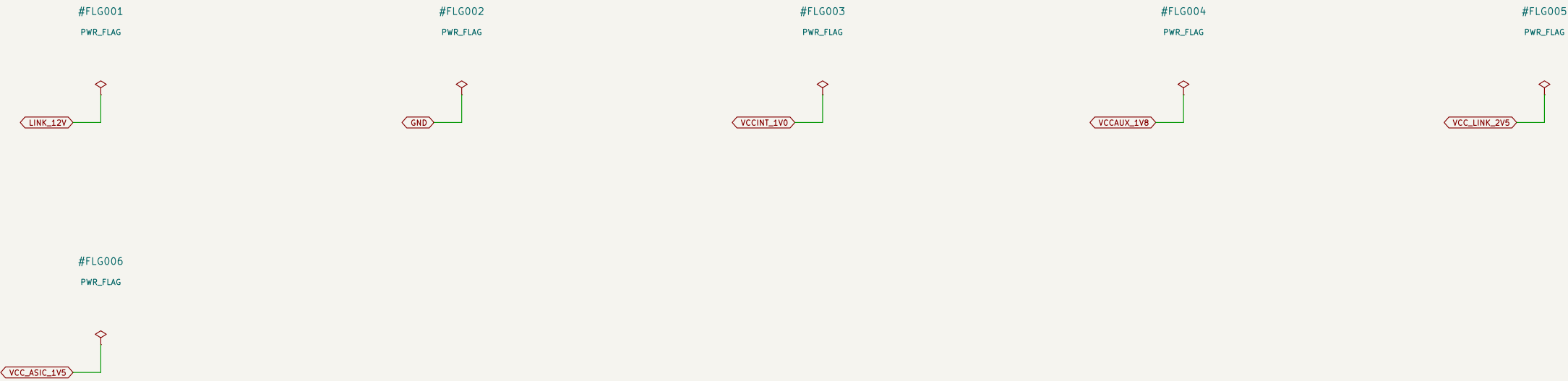
Reference clock

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



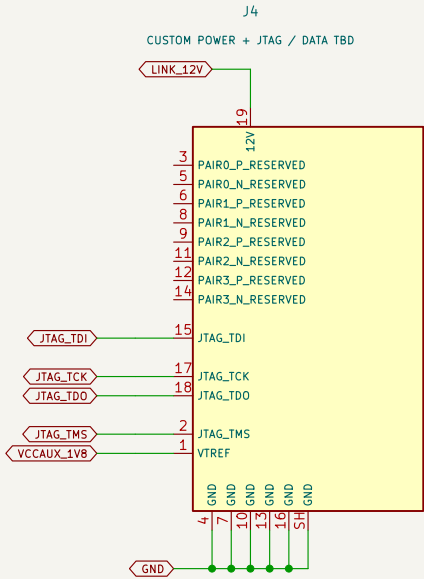
Power source assertions

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



Custom shared cable / power and JTAG

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



J4 / one custom cable connector

Contact	Net	Function
19	LINK_12V	Power input, proposed nominal 12 V
16	GND	DC return
1	VCC_LINK_2V5	Target voltage reference for programmer
2	JTAG_TMS	Programmer to FPGA E12
15	JTAG_TDI	Programmer to FPGA E11
17	JTAG_TCK	Programmer to FPGA E10
18	JTAG_TDO	FPGA E13 via R118 to programmer
4, 7, 10, 13, SH	GND	Pair shields and shell

Pins 3/5, 6/8, 9/11 and 12/14: reserved data pairs, not yet assigned.

12 V is a design proposal, not a qualified cable–power specification.

Requires a protected source and verified cable/contact current ratings.

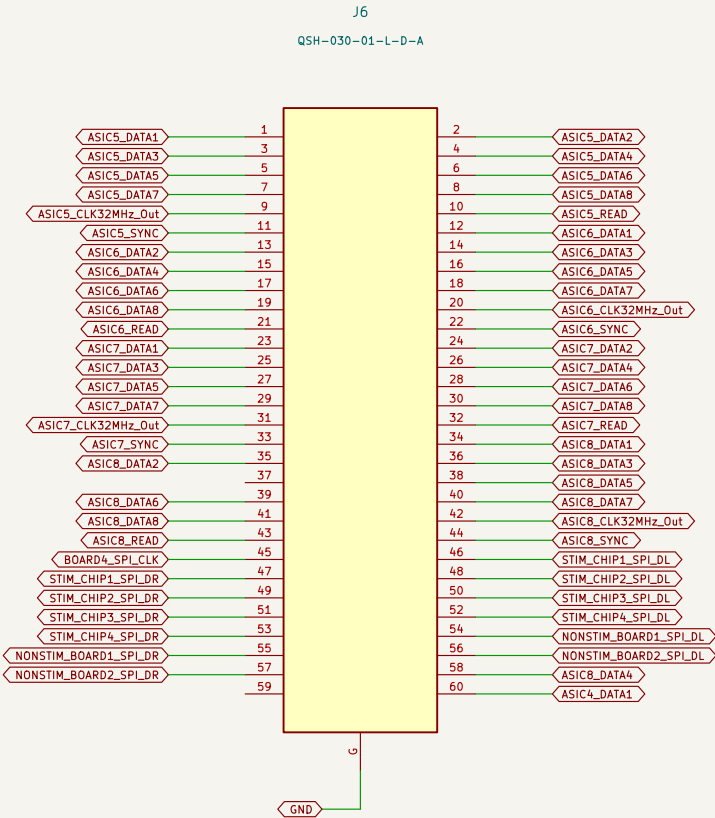
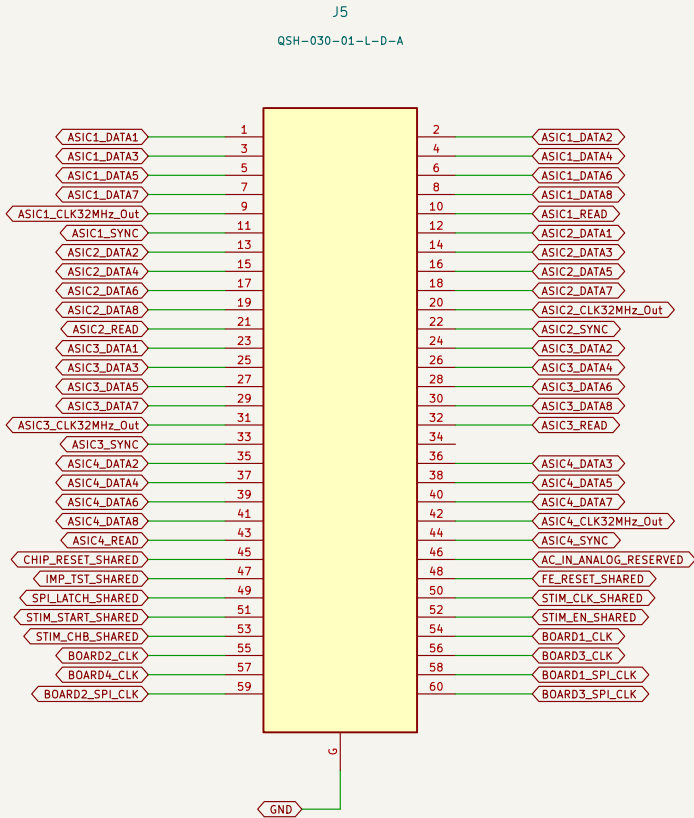
Do not connect this custom–powered port to ordinary HDMI equipment.

No separate JTAG header, boot–mode jumper or test points.

ASIC mezzanine /116 FPGA–linked signals +1 analog contact

117 contacts allocated:116 FPGA–linked signals and1 analog–reserved contact. J6.58–60 remain spare; ground blades are GND.
AC_IN analog0–1.5 V: source on routing/ASIC board pending. IMP_TST assumed digital1.5 V; inactive state pending. No ASIC supply on contacts.

J5.46: AC_IN analog0–1.5 V. Routing/ASIC–board analog source pending; no connection to FPGA.



No ASIC rail, analog reference or stimulation supply is allocated on these120 signal contacts.
Electrical thresholds, polarity, safe startup states, physical ASIC Identity and timing remain unapproved. These labels do not certify operation.